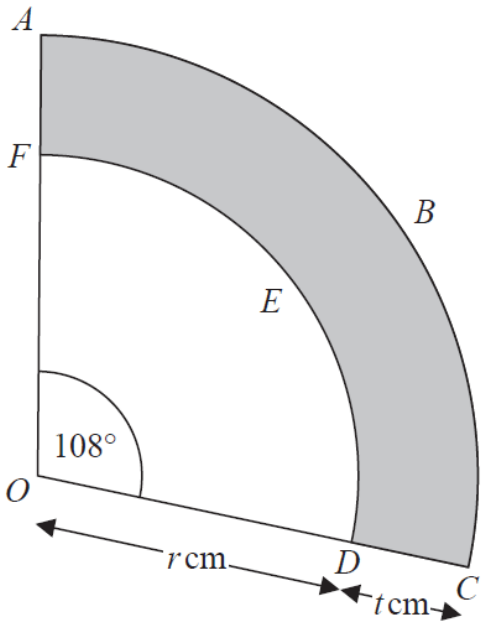
14



ABC and *DEF* are two arcs of circles, centre *O*.
OFA and *ODC* are straight lines.

(a) Show that the perimeter of the shaded region is given by $\frac{6\pi r + 3\pi t + 10t}{5}$ cm

Each arc = (108 ÷ 360) × 2πR = 3πR ÷ 5

1 mark
Use arc formula.

Outer R = r + t; inner R = r; straight sides = 2t.

$P = \frac{3\pi(r + t) + 3\pi r}{5} + 2t$

1 mark
Both arcs and sides.

$= \frac{3\pi(r + t) + 3\pi r + 10t}{5}$

1 mark
Common denominator.

$= \frac{3\pi r + 3\pi t + 3\pi r + 10t}{5}$

1 mark
Expand brackets.

$= \frac{6\pi r + 3\pi t + 10t}{5}$

1 mark
Collect like terms.

(5)

(b)(i) Find the exact value of $\dfrac{6\pi r + 3\pi t + 10t}{5}$ when $r = 0$ and $t = 10$

Give your answer in its simplest form.

$\dfrac{6\pi \times 0 + 3\pi \times 10 + 100}{5}$ $= 6\pi + 20$ cm

1 mark

Substitute r = 0 and t = 10.

.....

(1)

(ii) Explain what your value in part (b)(i) represents.

The perimeter of a sector of radius 10 cm

and angle 108°, including its two radii.

1 mark

The inner arc vanishes when r = 0.

(1)

(Total for Question 14 is 7 marks)